

CHAPTER # 1
STOICHIOMETRY

"CHEMISTRY"

DEFINITION:

"It is the branch of science which deals with the composition, properties and changes of matter. It also deals with the changes in matter and the principles which govern these changes."

"MATTER"

DEFINITION:

"Anything that has mass and occupies space is called matter."

"ATOM"

DEFINITION:

"The smallest particle of an element that can take part in a chemical reaction is called an atom."

EXAMPLE

Carbon, sodium, hydrogen, chlorine, copper, gold, iron etc.

"ELEMENT"

DEFINITION:

"Element is defined as a pure substance in which all the atoms have same chemical properties and have same atomic number."

EXAMPLE:

Carbon, sodium, hydrogen, chlorine, copper, gold, iron etc.

"MOLECULE"

DEFINITION:

"The smallest particle of a substance that can exist independently and shows the properties of original substance is called molecule."

EXAMPLE:

H₂O (water), N₂ (nitrogen), C₆H₁₂O₆ (glucose)

"ATOMIC NUMBER"

DEFINITION:

"The total numbers of protons or electrons present in an atom of an element is called atomic number."

EXAMPLE:

Atomic number of Sodium is 11. It has 11 protons and 11 electrons.

"MASS NUMBER"**DEFINITION:**

"The sum of number of protons and the number of neutrons present in the nucleus of an atom is called mass number."

EXAMPLE:

Mass number of Sodium is 23. It has 11 protons and 12 neutrons.

"ATOMIC MASS"**DEFINITION:**

"The mass of an atom of an element as compared to the mass of one atom of C^{12} is called atomic mass."

OR

"The average mass of natural mixture of isotopes which is compared to the mass of one atom of carbon-12 (most stable isotope of carbon)."

EXPLANATION:

- ✓ It is actually a ratio and hence has no units.
- ✓ Thus atomic masses may be given in any unit (grams, Kg, etc.).
- ✓ Generally atomic masses are expressed in atomic mass units (a.m.u.).
- ✓ One a.m.u. is defined as the mass equals to exactly one twelfth the mass of carbon-12 atom.

"FORMULA MASS"**DEFINITION:**

"The sum of atomic masses of all the atoms of different elements present in a compound is called formula mass."

EXAMPLE:

NaCl(58.5 a.m.u), $MgCl_2$ (95 a.m.u), $CaCl_2$ (111 a.m.u), MgO(40 a.m.u)

"GRAM ATOMIC MASS"**DEFINITION:**

"Atomic mass of an element expressed in grams is called gram atomic mass."

EXAMPLE:

Atomic mass of carbon=12 a.m.u

Gram atomic mass of carbon=12 gm

"GRAM FORMULA MASS"**DEFINITION:**

"Formula mass of a compound expressed in grams is called gram formula mass."

EXAMPLE:

Formula mass of NaCl=58.5 a.m.u

Gram formula mass of NaCl=58.5 gm

"MOLAR MASS"**DEFINITION:**

"The mass in grams of one mole of any pure substance is known as molar mass"

DEFINITION:"MOLAR VOLUME"

"The volume of one mole of a gas at standard temperature (273K) and pressure (1atm) is referred as molar volume"

DEFINITION:"MOLE"

"Atomic mass of an element or formula mass of a compound expressed in grams is called a mole."

OR

"The mass in grams of a substance that contains 6.02×10^{23} atoms, molecules or ions is called a mole."

Gram atomic mass = Mole

Gram formula mass = Mole

"AVOGADRO'S NUMBER"**DEFINITION:**

"One mole of any substance contains the same number of atoms, molecules or ions. It is known as Avogadro's number."

It is equal to 6.02×10^{23} .

EXAMPLES:

One mole of sodium contains 6.02×10^{23} atoms.

One mole of water contains 6.02×10^{23} molecules.

One mole of sodium chloride contains 6.02×10^{23} ions each.

"EXPONENTIAL NOTATIONS"**DEFINITION:**

"The short hand notation of very large or very small numbers is called exponential notation."

OR

"The method of expressing numbers as a factor of power 10 is called exponential notation."

"EXPONENT"**DEFINITION:**

"The power to the base 10 is called the exponent."

"LOGARTHME"**DEFINITION:**

"The expression $a^x = y$, x is called logarithm of y to the base "a", where "a" must be positive number other than one."

EXPLANATION:

In logarithm base is 10 so,

$$10^x = y \text{ (y is any real number)}$$

Logarithm is divided into two parts, the integer is called characteristic and the decimal fraction is called mantissa.

$$\text{Log } 273 = 2.4362 = x \text{ (with the help of log table or calculator)}$$

Characteristic = 2

Mantissa = 0.4362

The opposite of log is called antilog.

LOG RULES:

1. $\log ab = \log a + \log b$
2. $\log \frac{a}{b} = \log a - \log b$
3. $\log (a)^n = n \log a$

"LIMITING REACTANT"**DEFINITION:**

"A substance that produces least quantities of products after its complete consumption is called limiting reactant."

"STOICHIOMETRY"**DEFINITION:**

"The quantitative relationship between the reactants and the products in a balanced chemical equation is called stoichiometry."

"ERROR"**DEFINITION:**

"Error is defined as the difference or deviation in measured value and actual value."

FORMULA:

$$\% \text{ error} = \frac{\text{Actual value} - \text{measured value}}{\text{actual value}} * 100$$

SOURCES OF ERROR:

1. Defects in instruments
2. Improper handling of apparatus
3. Improper functioning of instrument(uncertainty)

TYPES:

- ✓ Positive or negative.
- ✓ Random or systematic.

"SYSTEMATIC ERROR"**DEFINITION:**

"The error which can be completely eliminated is called systemic error."

EXPLANATION:

Systematic or determinate errors are caused by the defect in the analytical method or by the improper functioning of instrument.

EXAMPLE:

In titration if the indicator is not properly prepared, then the color change will occur before the equivalence point.

"RANDOM ERROR"**DEFINITION:**

"The error which can be minimized but cannot be completely eliminated is called random error."

EXPLANATION:

Random or indeterminate errors are caused by the uncertainty in every physical measurement.

EXAMPLE:

During sudden change in temperature, change in humidity, fluctuation in potential difference (voltage).

It is an accidental error and is beyond the control of the person making measurement.

DEFINITION:**"EMPIRICAL FORMULA"**

"The formula which represents the simplest whole number ratio of different atoms present in a compound is called empirical formula."

EXAMPLES:

The empirical formula of glucose is CH_2O .

The empirical formula of benzene is CH .

DEFINITION:**"MOLECULAR FORMULA"**

"The formula which represents the actual number of atoms of different elements present in a molecule of a compound is called molecular formula."

EXAMPLES:

The molecular formula of glucose is $\text{C}_6\text{H}_{12}\text{O}_6$.

The molecular formula of benzene is C_6H_6 .

"SIGNIFICANT FIGURES"**DEFINITION:**

"The reliable digits in a number that are known with certainty are called significant figures".

(OR)

"The numbers which show confidence in measurements are called significant figures".

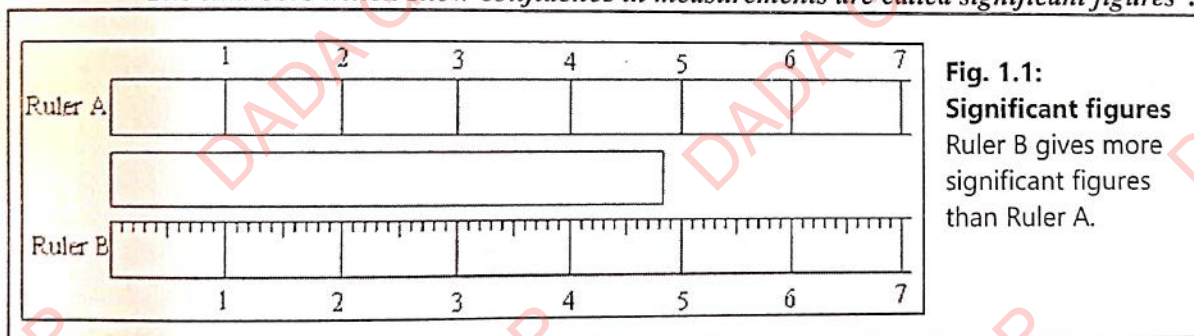


Fig. 1.1:
Significant figures
Ruler B gives more significant figures than Ruler A.

DETERMINATION OF SIGNIFICANT FIGURES:**RULES:**

1. All non-zero digits are significant.

Number Figures	Significant
73146	5

2. Zeroes that locate the decimal point in a number larger than one are not necessarily significant.

Number Figures	Significant
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70000	1
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3. Zero between the non-zero digits is significant.

Number Figures	Significant
70013	5

4. Zero locating the decimal point in a number less than one is not significant.

Number Figures	Significant
0.00245	3

5. Final zero to the right of the decimal point is significant.

Number Figures	Significant
73.000	5

"EMPIRICAL FORMULA"

DEFINITION:

"The formula which represents the simplest whole number ratio of different atoms present in compound is called empirical formula."

EXAMPLES:

The empirical formula of glucose is CH_2O

The empirical formula of benzene is CH

STEPS FOR DETERMINATION OF EMPIRICAL FORMULA:

The empirical formula of a compound is determined in the following way.

1. Detection of elements present in a compound.
2. Determination of masses of the elements.
3. Determination of percentages of the elements.
4. Determination of mole compositions.
5. Determination of simplest whole number ratio.

"MOLECULAR FORMULA"

DEFINITION:

The formula which represents the actual number of atoms of different elements present in molecule compound is called molecular formula.

EXAMPLES:

- ✓ The molecular formula of glucose is $\text{C}_6\text{H}_{12}\text{O}_6$
- ✓ The molecular formula of benzene is C_6H_6

DETERMINATION OF MOLECULAR FORMULA:

The molecular formula of a compound is determined in the following way.

1. Determination of empirical formula.
2. Determination of empirical formula mass.
3. Dividing the molecular mass by the empirical formula mass.
4. Multiplying the quotient of step (3) by the empirical formula.

THEORETICAL YIELD, PRACTICAL YIELD AND PERCENT YIELD:

There are three types of yields:

1. Theoretical yield
2. Percentage yield
3. Actual yield

"THEORETICAL YIELD"**DEFINITION:**

The maximum amount of product obtained by a balanced chemical reaction by using its Limiting reactant is known as Theoretical Yield.

"ACTUAL YIELD"**DEFINITION:**

The actual amount of product which is formed in experiment is called Practical or Actual yield.

"PERCENT YIELD"**DEFINITION:**

The efficiency of a chemical process is judged by calculating the ratio of practical yield to theoretical yield. This ratio is known as Percent yield.

$$\text{Practical Yield} = \frac{\text{Practical yield}}{\text{Theoretical yield}} \times 100$$

"ROUNDING OFF THE DATA"**DEFINITION:**

"Reducing of a number to the desired significant figures is called rounding off the data".

RULES:

1. If the digit to be dropped is greater than 5, the preceding digit is increased by one.
78.86 is rounded off to 78.9
2. If the digit to be dropped is less than 5, the preceding digit is left unchanged.
78.83 is rounded off to 78.8
3. If the digit to be dropped is equal to 5, then there are two possibilities;
 - If the final remaining digit is odd then add one to it.
78.835 is rounded off to 78.84
 - If the final remaining digit is even then it is left unchanged.
78.865 is rounded off to 78.86.

"DIFFERENCES"**ATOM AND MOLECULE:**

ATOM	MOLECULE
It is the smallest particle of an element.	It is the smallest particle of a substance.
It has only one nucleus.	It may have one or more nuclei.
It does not exist freely in nature (except the atoms of noble gases).	It exists freely in nature.

EMPIRICAL FORMULA AND MOLECULAR FORMULA:

EMPIRICAL FORMULA	MOLECULAR FORMULA
It represents the simplest whole number ratio of the atoms present in a compound.	It represents the actual number of atoms present in a molecule of a compound.
It represents both ionic and covalent compounds.	It represents covalent compounds only.
Example: The empirical formula of Benzene is CH	Example: The molecular formula of benzene is C ₆ H ₆

PHYSICAL AND CHEMICAL CHANGE:

PHYSICAL CHANGE	CHEMICAL CHANGE
It is the change during which composition of the matter does not change.	It is the change during which the composition of the matter changes.
It is a permanent change.	It is a permanent change.
In this change no new substance is formed.	In this change a new substance is formed.

DIFFERENCE BETWEEN ATOM AND ION

S.NO.	Atom	Ion
1.	The smallest particle of an element which can take part in a chemical reaction is called atom. An atom is electrically neutral because it contains equal number of positively charged particles (protons) and negatively charged particles (electrons).	An atom that carries an electric charge is called ion. . Thus electrically charged atoms are called ions which are formed by the loss or gain of electrons. In ion number of electrons and number of protons are not equal.
2.	With the exception of atoms of noble gases (e.g,ne,ar,kr,xen), all atoms have incomplete octet (an octet means 8 electrons in valence shell).	Ions usually possess complete instance Na ⁺ contains 8 electrons in its valence shell similarly Cl ⁻ also has 8 electrons in its valence shell.
3.	Na,Ca,S,C,Cle etc are the symbols of some atoms.	Ions are of two types:- (a)Cation, formed by loss election e.g.Na ⁺ (b)Anion formed by gain of electron e.g.Cl ⁻

DIFFERENCE BETWEEN SYSTEMATIC ERRORS AND RANDOM ERROR

Systematic Errors	Random Errors
Systematic or determinate errors are caused by the defect in the analytical method or by the improper functioning of instrument.	Random or indeterminate errors are caused by the uncertainty in every physical measurement.
Systematic errors are avoidable because the cause of error may be eliminated or corrected (with in our control).	Random errors are unavoidable as the cause of such errors cannot be

Easy to predict	eliminated completely (beyond human control).
DIFFERENCE BETWEEN GRAM ATOM AND GRAM MOLECULE	
	Difficult to predict

S.No.	Gram Atomic Mass (Gram Atom)	Gram Molecular Mass (Gram Molecule)
1.	The relative atomic mass of an element expressed in gram is called gram atomic mass or gram atom.	The relative molecular mass of a substance (element compound) expressed in gram is called gram molecular mass or gram molecule.
2.	The one-gram atom of all elements contains 6.20×10^{23} atom.	The one gram molecule of all compounds contains 6.02×10^{23} molecules.
	Examples:	Examples:
(i)	1 gram atom of H = 1gm	1 gram molecule of H_2 = 2gm
(ii)	1 gram atom of Na = 23gm	1 gram molecule of H_2SO_4 = 98gm
(iii)	1 gram atom of O = 16gm	1 gram molecule of O_2 = 32gm
(iv)	1 gram atom of Cl = 35.5gm	1 gram molecule of Cl_2 = 71gm

"OBJECTIVES"**FILL IN THE BLANKS:**

- 1 mole of a gas at S.T.P occupies a volume of 22.4 dm^3 .
- A gas occupying a volume of 22.4 dm^3 at S.T.P contains 6.02×10^{23} molecules.
- A formula, which gives the relative number of atoms in the molecule of a compound, is called empirical formula.
- A formula that gives the actual number of all kinds of atoms present in the molecule of compound is termed as molecular formula.
- The chemical formula that not only gives the actual number of atoms but also shows the arrangement of different atoms present in the molecule is called structural formula.
- Atomic mass or molecular mass expressed in grams is known as mole.
- 2 moles of H_2O contain 36 grams and 1.204×10^{23} numbers of molecules.
- 1 mole of NaOH is equal to 40 gm of NaOH.
- The S.I unit of a mass is kilogram.

10. A ton mole of iron is equal to 56 tons.
11. For three significant figures, 25.55 is rounded off to 25.6.
12. Mole is the quantity, which has 6.02×10^{23} particle of the substance.
13. In rounding off right most figures is dropped.
14. The atomic number is equal to the number of protons in nucleus.
15. Very small and very large quantities are expressed in terms of exponents.
16. The average weight of atoms of an element as compared to the weight of one atom of C_{12} is called the atomic mass.
17. 1.0007 contains 5 significant figures.
18. The figure 24.75 will be rounded off to 24.8.
19. Empirical formula weight is the sum of atomic weights of the elements represented by the Empirical formula of the compound.
20. The sum of atomic weights of all the elements present in molecular formula is called the molecular weight.
21. The degree of accuracy of a measured quantity increases with increasing number of significant figures in it.
22. The atomic mass of sodium is 23 a.m. u.
23. The symbolic representation of a molecule of a compound is called formula.
24. Molecular formula of chloroform is $CHCl_3$ and its Empirical formula is $CHCl_3$.
25. Molecular formula of benzene is C_6H_6 and its empirical formula is CH .
26. 58.5 is the formula mass of $NaCl$.
27. 4.5 gm of nitrogen will have 9.6×10^{22} molecules.
28. The accuracy of measurement depends on the number of significant figures.
29. 2 moles of SO_2 is equal to 128 gm.
30. 1000 gm of H_2O is equal to 55.55 moles.
31. 6.02×10^{23} is called the Avogadro's number.
32. Mole is the amount of substance, which contains as many numbers of particles as there are 12 gm of Carbon.

33. 0.0087 has two significant figures
34. 0.0006 has one significant figure
35. 7.40×10^8 has two significant figures.
36. 7×10^8 has one significant figure.
37. Usually Molecular formula is simple multiple of the empirical formula.
38. 0.1 mole of H_2O contains molecules of H_2O 6.02×10^{22} .
39. Mass of 3.01×10^{22} molecules of CO_2 is 2.2 grams.
40. 4.5 gm of water will have 1.505×10^{23} molecules.

TRUE FALSE

1. One gram mole of oxygen gas, O_2 is equal to 16 gm. (False)
2. A gram molecular weight of H_2 contains the same number of molecules as a gram molecular weight of Cl_2 . (True)
3. Volume of 1 ounce mole of a gas at S.T.P is equal to 22.4 Cubic feet. (True)
4. Empirical Formula always differs from Molecular formula. (false)
5. An empirical formula may be the same as the molecular formula. (True)
6. 342 gms of cane sugar, $C_{12}H_{22}O_{11}$ consists of 6.02×10^{23} molecules. (True)
7. Weight of an atom of Hydrogen is 1.008 gm. (False)
8. The numbers in which last digit is in doubt is called significant figure. (True)
9. Molecular formula is simple multiple of the empirical formula. (True)
10. A formula unit represents Empirical Formula. (True)
11. In rounding off first figure is dropped. (False)
12. .Mole is that quantity which has 6.02×10^{23} particles of the substance. (True)
13. 0.00002 has five significant figures. (False)
14. Zero becomes significant when it is placed between significant figures. (True)
15. Zeros at the end of a number and to the right of decimal point are not significant. (False)
16. One gram mole of a gas at S.T.P has a volume of 22.4 Cubic feet. (False)

17. Exponential form of a number consists of a coefficient multiplied by a power of 10. (True)
18. Significant figures are those, which serve to establish the exact, value of the number when expressed as a measurement. (True)
19. An empirical formula is that which gives exact number of each kind of atoms present in a molecule of a compound. (False)

"NUMERICALS"

SIGNIFICANT FIGURES:

- Using significant figure rules simplify:
$$\frac{56 \cdot 725 \cdot 273}{760 \cdot 298}$$
- Simplify according to the rules of significant figure:
$$\frac{2.147 \cdot 8.123}{4.956}$$
- Multiply $4.46 \cdot 1.2$ according to multiplication rules.
- How many significant figures are there in 0.0821?
- Give the significant values of: 685, 9000, 0.58, 0.004
- How many significant figures are there in 732.020? Convert it into three and two significant figures.
- Solve using the rules of significant figures: $3.94 \cdot 2.12345$
- Solve according to the rules of significant figures:
$$\frac{7.856 \cdot 2.5}{3.35}$$
- Simplify according to the rules of significant figures : $43100 + 3900 + 2100$
- The mass of a substance is 18.8865 gram and its volume is 7.9 cm^3 . What will be its density considering significant figures and rounding off the above obtained.

EXPONENTIALS /SCIENTIFIC NOTATIONS/LOGARITHMS:

- Simplify the following by using exponential notation: $43100 + 3900 + 2100$.
- Express in power of 10:
 - 50000
 - 53.24
 - 0.000005
 - 54321
- Express in power of 10:
 - 5678
 - 432.1
 - 2000
 - 0.00012

BASIC CALCULATIONS:

- Find the mass and number of molecules in $18000\text{cm}^3\text{H}_2\text{S}$ gas at S.T.P.
27.32gm, 4.83×10^{23} molecules
- In a collection of 24×10^{25} molecules of $\text{C}_2\text{H}_5\text{OH}$, what is the number of moles?
398.67 moles
- What is the mass of 3.01×10^{22} molecules of N_2 .
1.4gm
- The atomic mass of Na is 23 amu;
 - Calculate the mass of 2.408×10^{23} atoms of Na.
9.2 gm
 - Calculate the number of moles of Na in 4.6 gm.
0.2 moles
- Calculate the number of moles in 3800g of:
 - $\text{Ca}(\text{OH})_2$
51.35 moles
 - SO_2
59.37 moles
 - H_2
1900 moles
- Calculate mass in grams of 3.01×10^{23} molecules of water.
9gm
- Calculate the no of moles, the no of molecules and the volume in cm^3 of 0.32gm of:
 - CH_4
0.02 mole, 1.204×10^{22} molecules, 448cm^3
 - And SO_2 gases at S.T.P
0.005 mole, 3.01×10^{21} molecules, 112cm^3
 - Calculate the number of moles and number of molecules present in 9gm of $\text{C}_6\text{H}_{12}\text{O}_6$.
0.05 mole, 3.01×10^{22} molecules
- Atomic mass of sodium is 23amu, calculate:
 - The no of moles in 400gm of Na.
17.39 mole
 - The no of atoms in 4.6gm of Na.
 1.2×10^{23} atoms
 - Mass of 3.01×10^{23} molecules of Na.
0.0115gm
- The molecular weight of NaOH is 40, find;
 - Its gram molecular weight(moles)
40gm
 - Weight of 3 molecules in amu.
120 amu
 - Weight of 60200 molecules in gram.
 $4 \times 10^{-18}\text{gm}$
 - Weight of 6 gram-moles.
240 gm
- The atomic mass of Zinc is 65.4 amu. calculate ;
 - The number of moles and also the number of atoms in 10.9 gm of zinc.
0.166 mole, 1.00×10^{23} atoms
 - The mass of 1.204×10^{24} atoms of zinc in gm.
130.8 gm
- A sample of chlorine gas at S.T.P has volume of 800cm^3 . calculate;
 - The number of moles of chlorine.
0.0357 mole
 - The mass of the sample.
2.5gm
 - The number of chlorine molecule in the sample.
 22.35×10^{23} molecules
- How many atoms of C are present in 10 grams of coke?

STOICHIOMETRY:

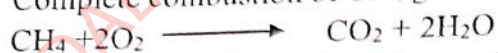
1. 54 gm of solid dinitrogen pentaoxide, N_2O_5 is decomposed on heating as;



Find the volumes of NO_2 and O_2 at S.T.P.

$$NO_2 = 22.4 dm^3, O_2 = 5.6 dm^3$$

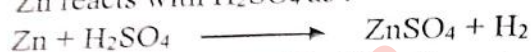
2. Complete combustion of CH_4 gives the reaction:



calculate the mass and the volume of CO_2 gas produced at S.T.P by the combustion of 9.6 gm of CH_4 .

$$26.4 \text{ gm } CO_2, 13.02 \text{ dm}^3 CO_2$$

3. Zn reacts with H_2SO_4 as :



calculate the mass of $ZnSO_4$, the volume of H_2 and the number of molecules of H_2 which will be produced by reacting 163.5 gm of Zn with H_2SO_4 at S.T.P. (atomic mass of Zn = 65.4)

$$403.5 \text{ gm } ZnSO_4, 56 dm^3 H_2, 1.505 \times 10^{24} \text{ molecules } H_2$$

4. Find the mass of $KClO_3$ required to produce 3.2 gm of O_2 ;



$$8.16 \text{ gm } KClO_3$$

5. 100 gm of KNO_3 are heated to redness. What volume of oxygen at $39^\circ C$ and 765 mm pressure will evolve?



$$12.58 \text{ dm}^3 O_2$$

6. 30 gm of limestone ($CaCO_3$) was heated:



a. Write the equation for the reaction.

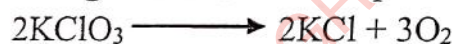
b. Calculate the mass of CO_2 produced.

$$13.2 \text{ gm } CO_2$$

c. Calculate the volume of CO_2 at S.T.P

$$6.72 \text{ dm}^3$$

7. 73.5 gm $KClO_3$ is decomposed on heating as follows:



calculate;

a. The mass of KCl .

$$44.7 \text{ gm}$$

b. The volume of O_2 formed at S.T.P.

$$20.16 \text{ dm}^3$$

8. Find the mass, number of moles, number of molecules and volumes of carbon dioxide gas at STP obtained by burning 6 gm of carbon. Give also the equation.

$$22 \text{ gm}, 0.5 \text{ mol}, 3.01 \times 10^{23}, 11.2 \text{ dm}^3$$

9. Calculate the volume of O_2 at STP required to burn 30 gm of CH_4 gas as shown in the reaction



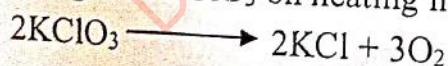
$$42 \text{ dm}^3$$

10. Calculate the volume of nitrogen gas produced by heating 800 gm of ammonia at $21^\circ C$ and 5 torr pressure.



$$525.86 \text{ dm}^3$$

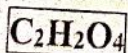
11. Calculate the volume of oxygen gas at STP that may be obtained by complete decomposition of 51.3 gm of $KClO_3$ on heating in the presence of MnO_2 as a catalyst;



$$14.07 \text{ dm}^3$$

EMPIRICAL FORMULA AND MOLECULAR FORMULA:

1. An organic compound carbon, hydrogen and oxygen has 26.7% C, 2.2% H by weight. Find the molecular formula of the compound, if the molecular mass of the compound is 90 amu.



- What is the molecular formula of the compound that contains 80% carbon and 20 % hydrogen? Its molecular mass is 30 amu.
- An organic compound producing air pollution contains 8.73% carbon, 77.45% chlorine and 13.82% fluorine. Find the molecular formula of the compound if the molecular mass is 137.5 amu. (atomic mass C=12, Cl=35.5, F=19) C_2H_6
- Di-ethyl Zinc is a chemical used in library to protect books from worms; its composition is 53% Zinc, 38.9% carbon and 8.1% hydrogen. Find the empirical formula of the compound. C_2Cl_3F
 ZnC_4H_{10}
- An organic compound contains 67.7% carbon, 7.41% hydrogen and 25.9% nitrogen. The molecular mass of the compound is 108. Determine the empirical and molecular formula of the compound.
- The molecular mass of compound is 180 and its empirical formula is CH_2O . Find the molecular formula. C_3H_4N , $C_6H_8N_2$
- Acetic acid contains C, H and O. 2.14 gm of a sample of Acetic acid on complete combustion gave 3.105 gm of CO_2 and 1.27 gm of H_2O . The molecular mass of Acetic acid is 60. Find its empirical and molecular formula. $C_6H_{12}O_6$
- A hydrocarbon has 20% H and 80% C. Find the empirical and molecular formula of the compound if the molecular mass is 30. CH_2O , $C_2H_4O_2$
- An organic compound contains C=51.80%, H=13.12% and O=35.8%, determine the empirical formula of the compound. CH_3 , C_2H_6
- Adipic acid is used in the manufacture of Nylon. The acid contains 43.9% C, 6.9% H, and 43.8% O by mass. The molecular mass of the acid is 146 amu; find the molecular formula of Adipic acid. C_2H_6O
- A given compound contains 75.2% carbon, 10.75% hydrogen and 14.05 % oxygen; calculate the empirical formula of the compound. $C_3H_5O_2$, $C_6H_{10}O_4$
- 1.0 gm of a sample of an organic compound was burnt in excess of oxygen to yield 3.03 gm of CO_2 and 1.55 gm of H_2O ; if the molecular mass of the compound is 58, find its molecular formula. $C_7H_{12}O$
- Combustion of 0.5 gm of hydrogen produces 1.515 gm CO_2 and 0.77 gm of H_2O . If the molecular mass of the compound is 58 amu. Determine its molecular formula. C_4H_{10}

LIMITING REACTANT:

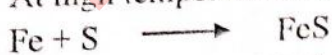
- Find the weight of Barium sulphate precipitated on adding a solution containing 82 grams of potassium sulphate to a solution containing 82 grams of Barium chloride.
 $K_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2KCl$
- What is the minimum mass of $Al(OH)_3$ that can be obtained by the reaction of 13.4 gm of $AlCl_3$ with 10g of NaOH according to the following equation;
 $AlCl_3 + 3NaOH \longrightarrow Al(OH)_3 + 3NaCl$ $6.5\text{gm } Al(OH)_3$
- For the reaction; $3Mg + N_2 \longrightarrow Mg_3N_2$
1.5 gm of magnesium and 1.5g of nitrogen react together. What is the actual amount of Mg_3N_2 formed and which element is the limiting reactant? $2.08\text{gm } Mg_3N_2$, Mg is L.R
- You are provided with 6g of C and 100g of O_2 . Calculate the amount of CO_2 prepared by reacting them. Which of them is limiting reactant? 22gm C , C is L.R

Chemistry I

"STOICHIOMETRY"

Chapter # 1

5. At high temperature sulphur combines with iron to form brown-black iron sulphide.



In one experiment 76.2gm of Fe are allowed to react with 86.7g of S. Which of the two is limiting reactant? Calculate the mass of FeS obtained. (atomic mass Fe=56, S=32)

Fe is L.R, 119.74 gm FeS

6. ZnCl_2 is prepared by the reaction $\text{Zn} + 2\text{HCl} \longrightarrow \text{ZnCl}_2 + \text{H}_2$
6.54 gm of Zn reacts with 73 gm of HCl. Find the limiting reactant and the mass of ZnCl_2 produced.

Zn is L.R, 13.64gm ZnCl_2

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